

# **EXPLORATORY DATA ANALYSIS USING POWER BI**

**WEEK 6 - REPORT  
HEALTHCARE VISUALIZATIONS AND DASHBOARD**

**Submitted By  
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## Introduction:

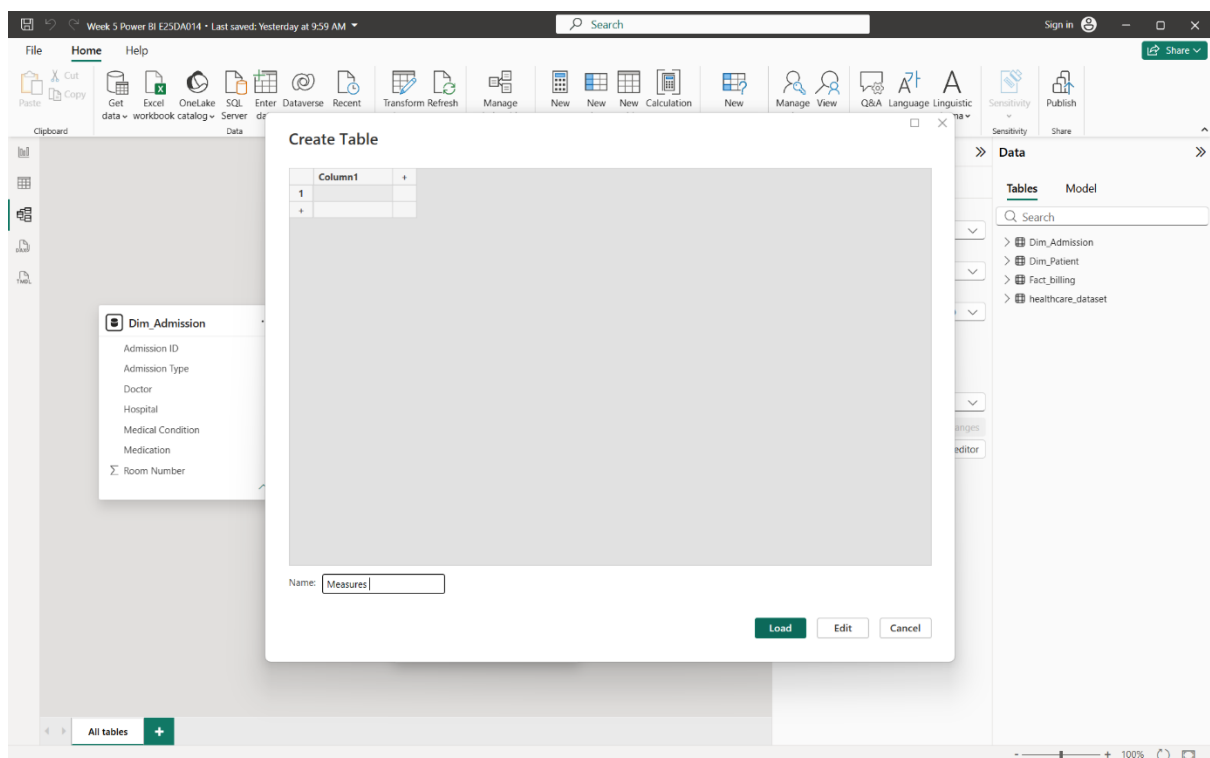
As the continuation of Week 5, after completing data cleaning and data modeling, DAX measures were created to calculate important healthcare metrics. These measures were then used to construct the visual layout and interactive dashboard in Power BI.

## Objectives:

- To create DAX measures for important healthcare KPIs.
- To calculate patient count, billing amount, average billing, and average length of stay.
- To create appropriate visualizations for healthcare analysis.
- To design an interactive dashboard using slicers and cross-filtering.

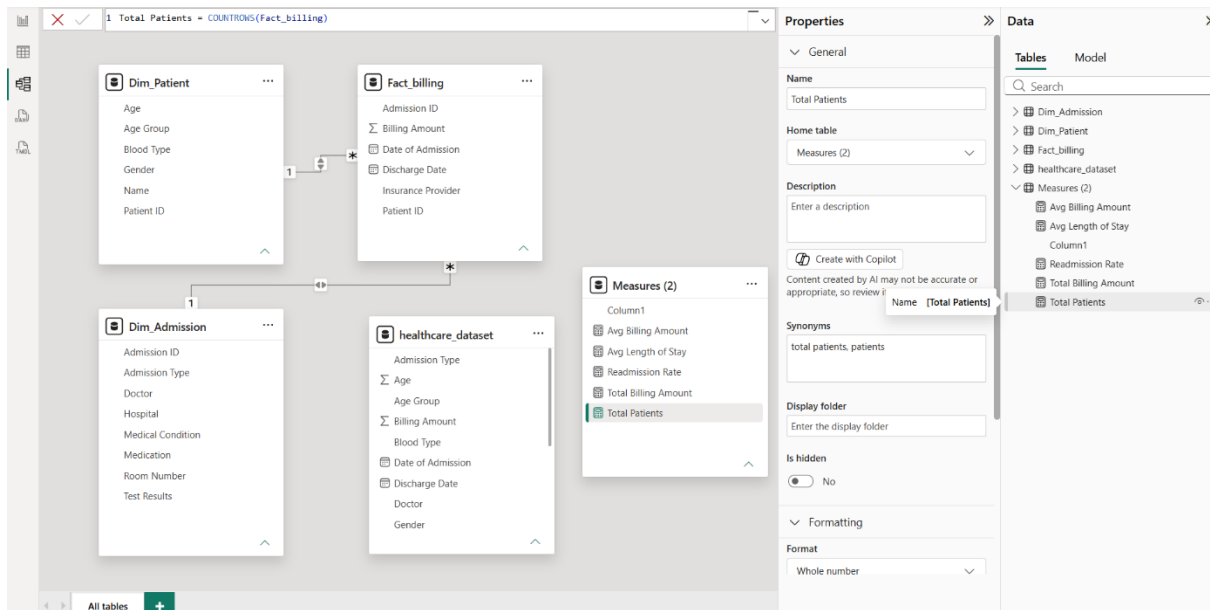
## 1. DAX Measures Created:

### 1.1 Measures Table Created:



### 1.2 Total Patients

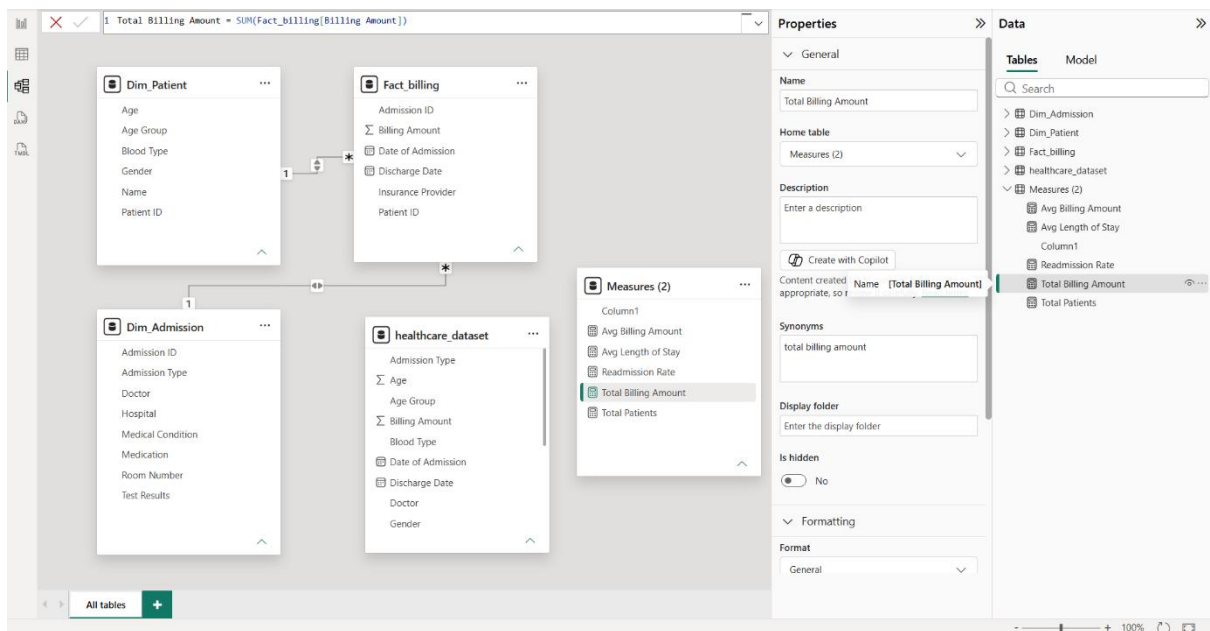
Total Patients = COUNTROWS(Fact\_Billing)



### 1.3 Total Billing Amount

Total Billing Amount =

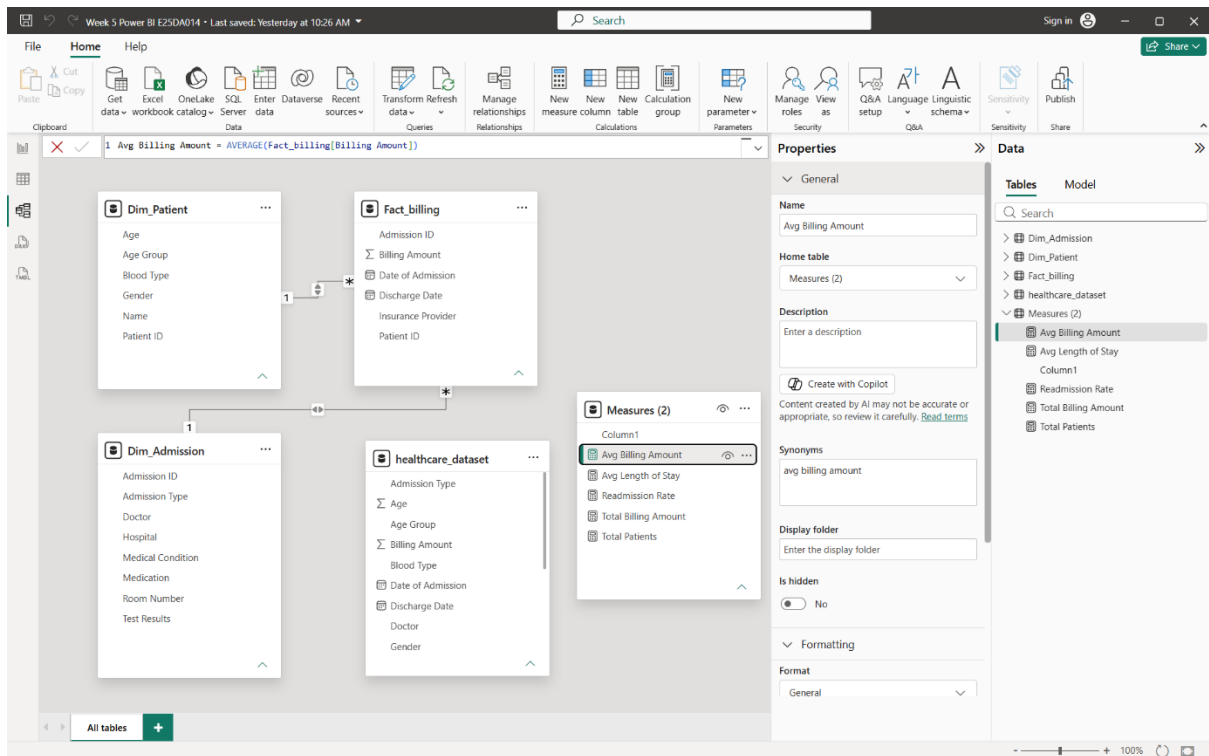
`SUM(Fact_Billing[Billing Amount])`



### 1.4 Average Billing Amount

AVG Billing Amount =

`AVERAGE(Fact_Billing[Billing Amount])`



## 1.5 Average Length of Stay

AVG Length of Stay =

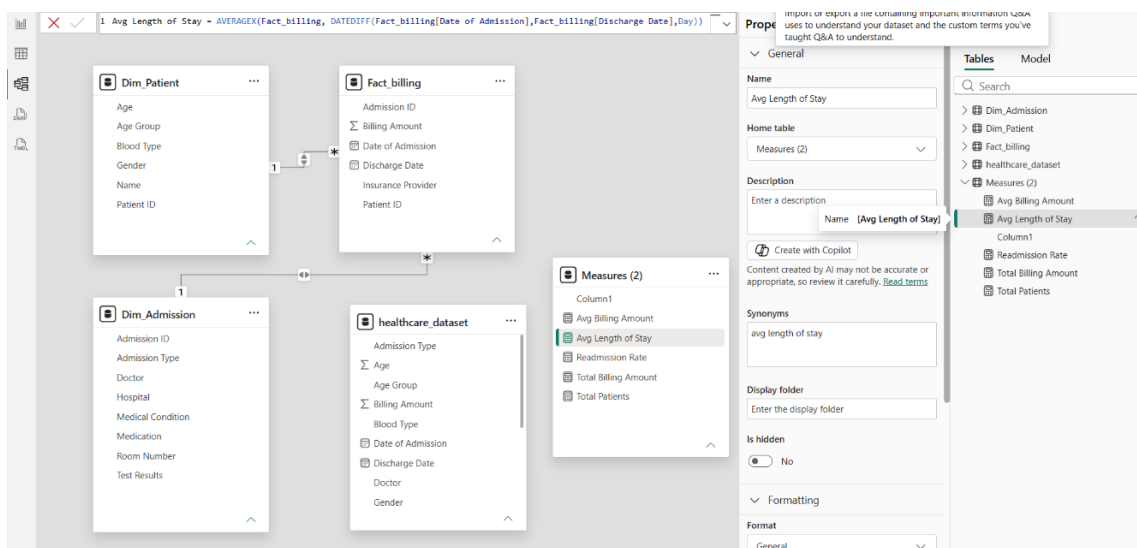
AVERAGEX( Fact\_Billing,

DATEDIFF(

Fact\_Billing[Date of Admission],

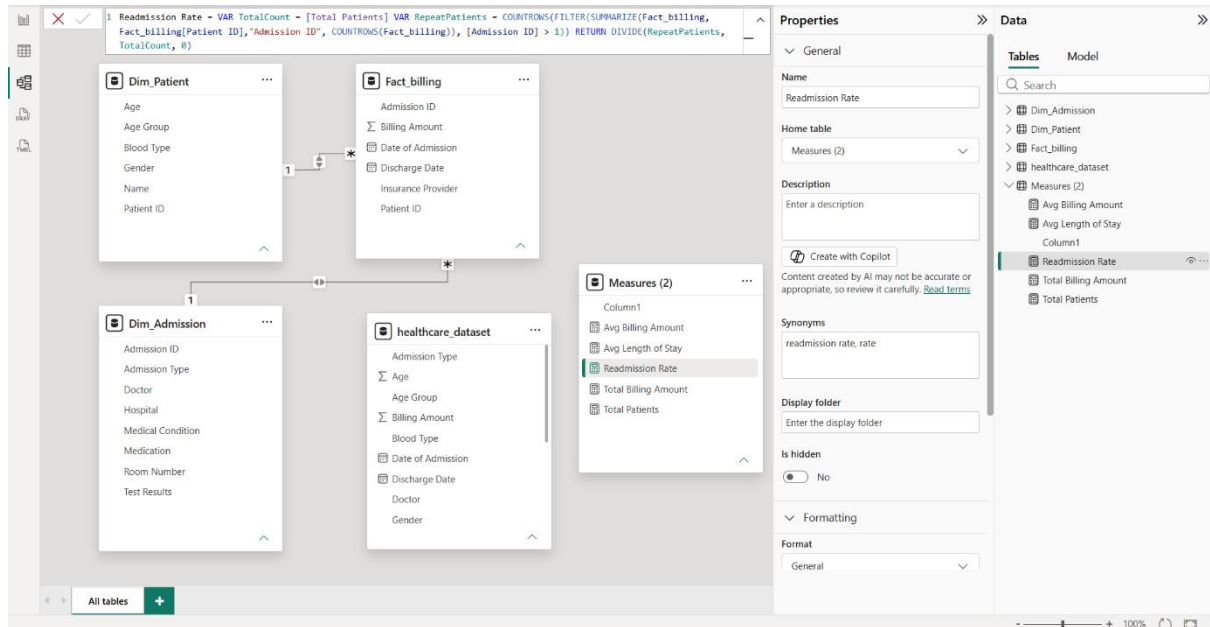
Fact\_Billing[Discharge Date],

DAY) )



## 1.6 Readmission Rate

A DAX measure was also created to identify patients with more than one admission and calculate the readmission rate.



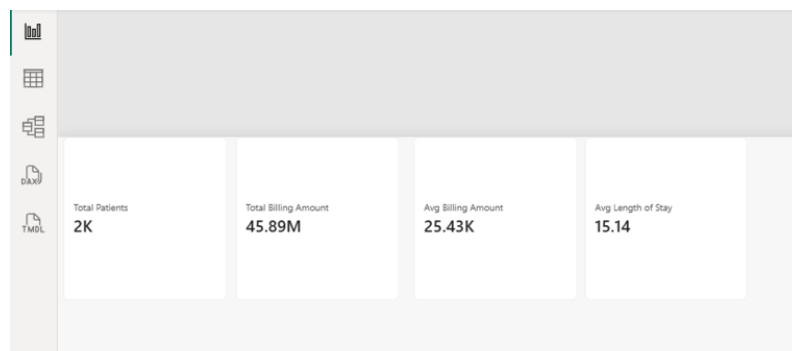
## 2. Constructing Visual Layout & Dashboard Elements

### 2.1 Top KPI Summary Cards

Four **Card visuals** were created at the top of the dashboard to display the main healthcare KPIs:

- Total Patients
- Total Billing Amount
- Average Billing Amount
- Average Length of Stay

These cards provide a quick summary of the overall healthcare data.

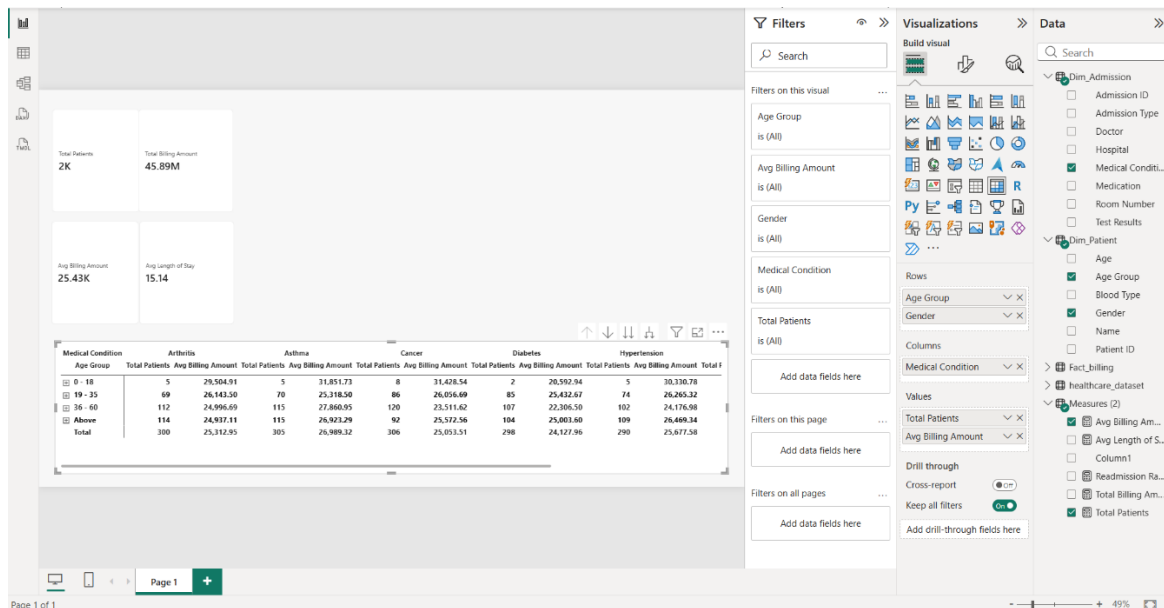


## 2.2 Demographics & Patient Distribution

### Matrix Visual:

A **Matrix visual** was created with:

- **Rows:** Age Group, Gender
- **Columns:** Medical Condition
- **Values:** Total Patients, Average Billing Amount

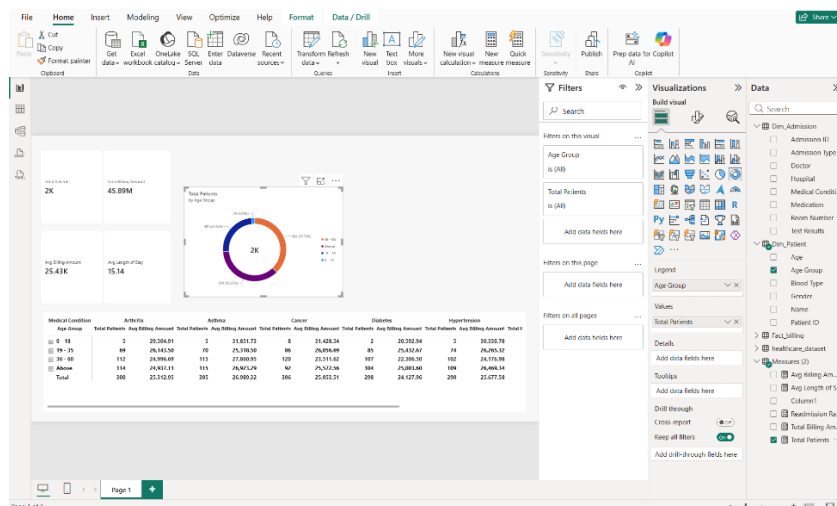


This visual helps analyse patient distribution and billing across different age groups, genders, and medical conditions.

### Donut Chart:

A **Donut Chart** was created with:

- **Legend:** Age Group
- **Values:** Total Patients



This shows the distribution of patients across different age groups.

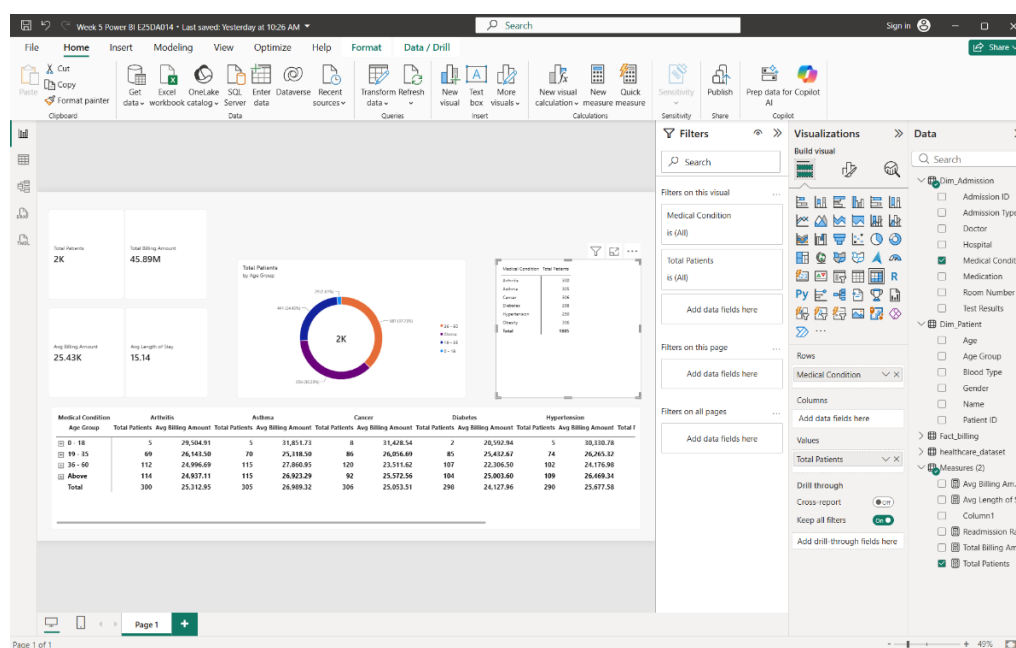
## 2.3 Medical Conditions & Admission Volumes

### Matrix Visual:

A visualization was created to analyze patient admissions based on medical conditions and admission types.

- **Rows:** Medical Condition
- **Values:** Total Patients

This helps compare the number of patients across different medical conditions while also showing the distribution by admission type.



## 2.4 Admission Volume Trends & Hospital Analytics

### Line Chart:

A Line Chart was created to display admission trends over time.

- **X-Axis:** Date of Admission — Year, Month
- **Y-Axis:** Total Patients, Total Billing Amount

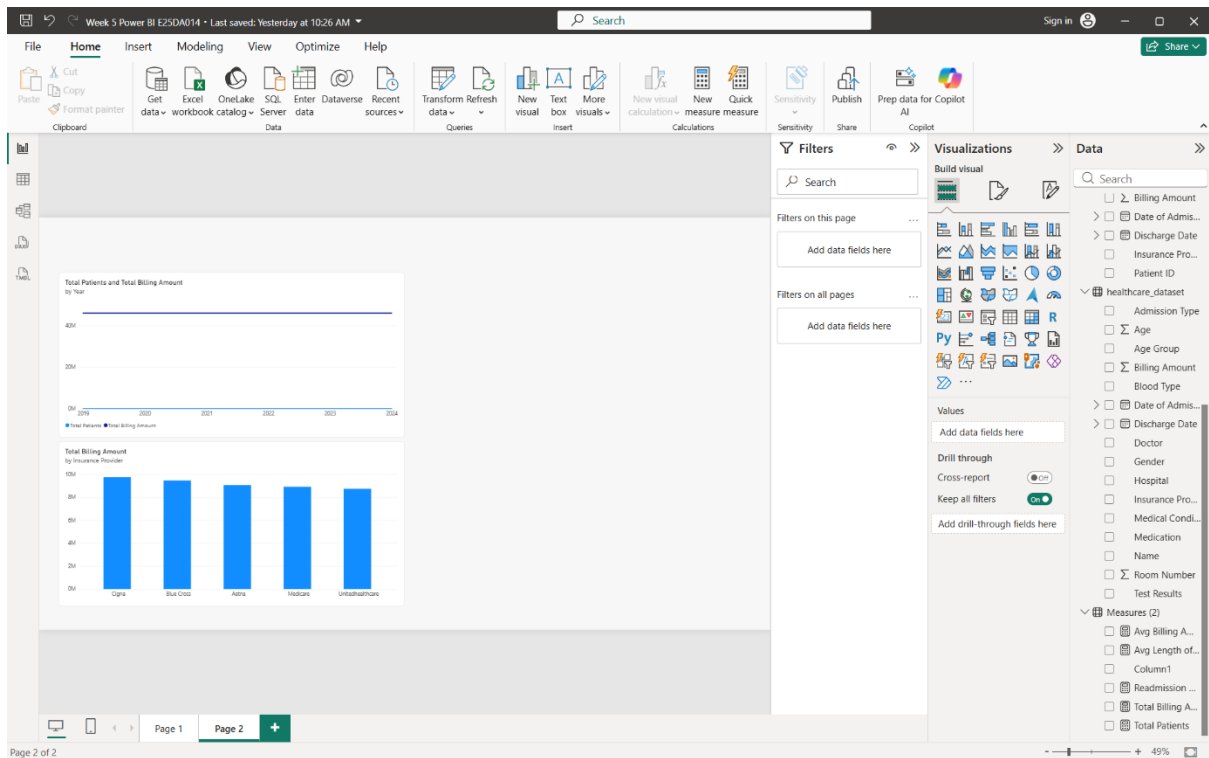
This visualization helps observe changes in patient admissions and billing amounts over time.

### Clustered Column Chart:

A Clustered Column Chart was created to compare billing amounts among insurance providers.

- **X-Axis:** Insurance Provider
- **Y-Axis:** Total Billing Amount

This allows comparison of the total billing amount associated with different insurance providers.



## 2.5 Slicers & Dynamic Cross-Filtering

Three slicers were added to make the dashboard interactive.

### Slicer 1 – Insurance Provider

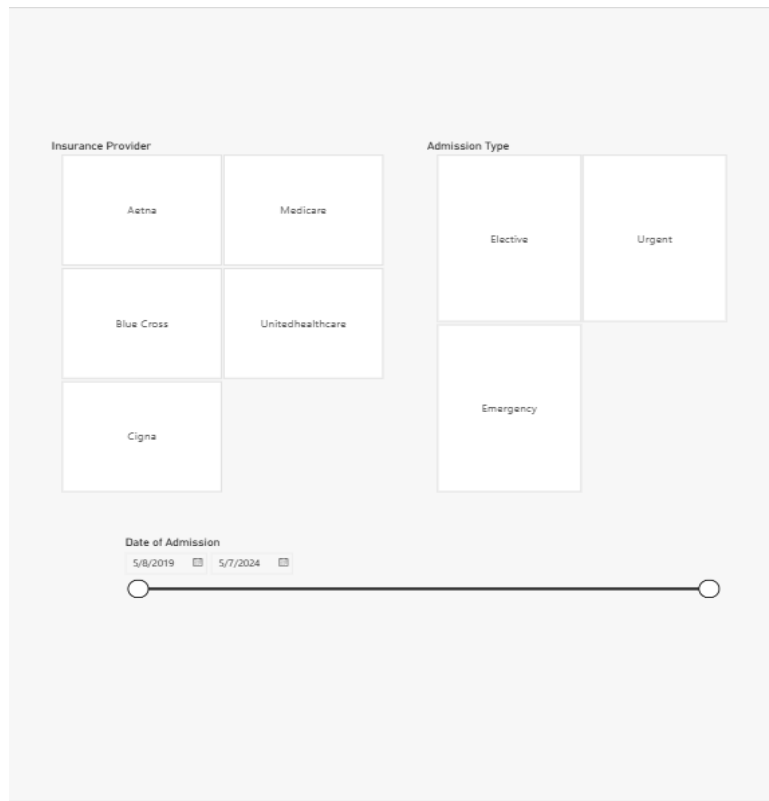
- **Field:** Insurance Provider
- **Type:** Tile

### Slicer 2 – Admission Type

- **Field:** Admission Type
- **Type:** Tile

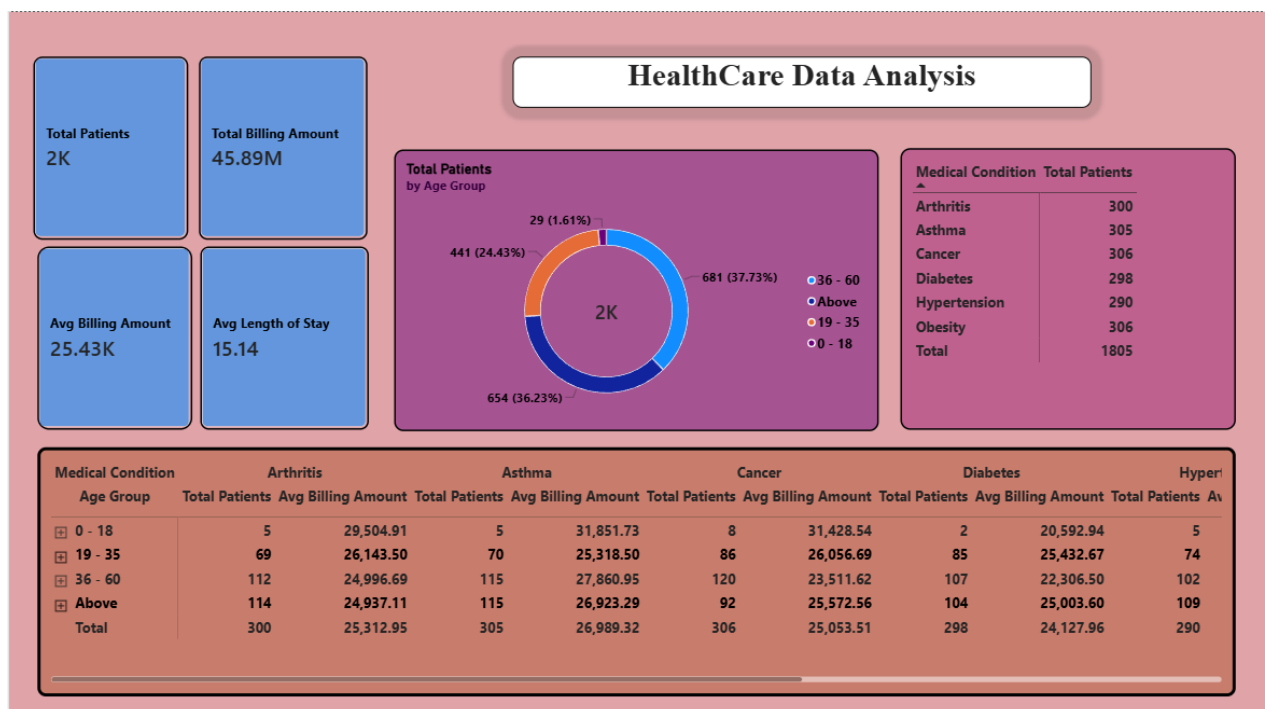
### Slicer 3 – Date Range

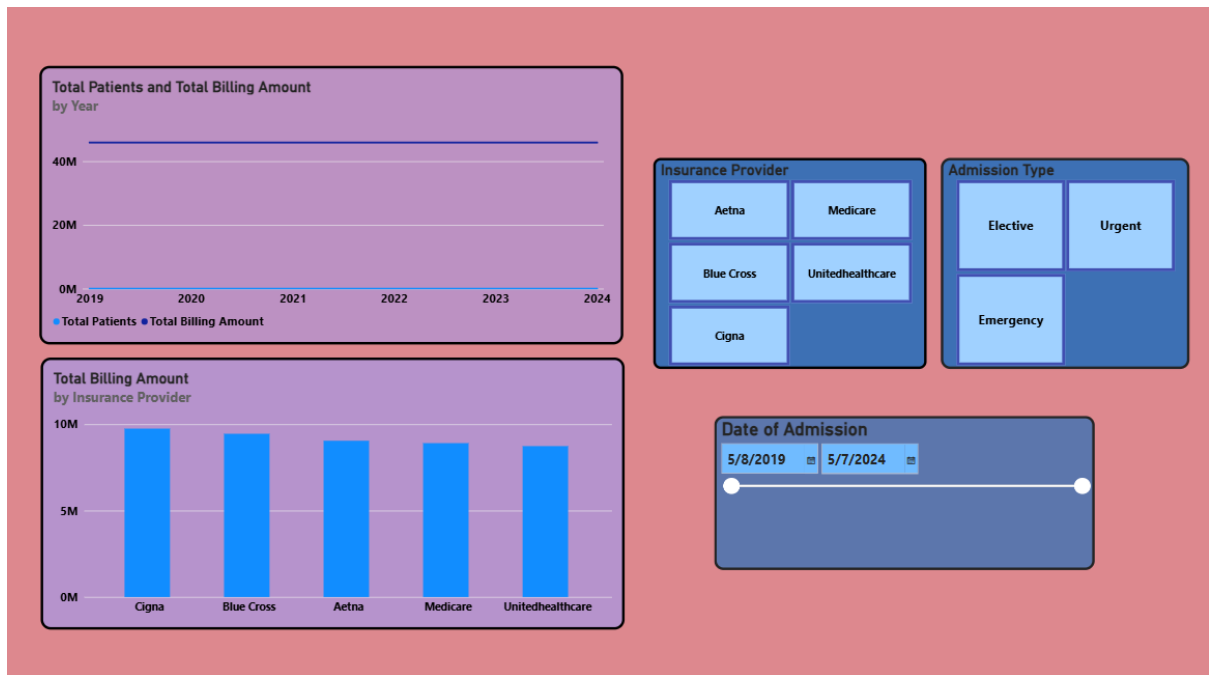
- **Field:** Date of Admission
- **Type:** Date Range



These slicers allow users to filter the dashboard dynamically based on insurance provider, admission type, and admission date.

### Final Dashboard Layout:





## Conclusion:

The cleaned and modeled healthcare dataset was converted into an interactive Power BI dashboard. DAX measures were created for important healthcare KPIs, followed by the creation of cards, matrix, donut chart, stacked bar chart/matrix, line chart, clustered column chart, and slicers. These dashboard elements provide an interactive way to analyse **patients, medical conditions, admissions, billing, insurance providers, and admission trends**.